

2025 Ch H1 Q9

Section: Nature's Chemistry

Topic: Oxidation of Food

Question summary

Identify the correct structural formula for 6-methylhept-5-ene-2-one.

Worked solution

Parent chain: hept -> 7 carbons.

Suffix -one at position 2 -> carbonyl (C=O) group on carbon 2.

Double bond at position 5 -> C=C between carbons 5 and 6.

Substituent 6-methyl -> CH₃ branch on carbon 6.

Combined structure: CH₃COCH₂CH₂CH=C(CH₃)CH₃, which matches option C.

Final answer

C. CH₃C(CH₃)CHCH₂CH₂COCH₃

Revision tips

- The suffix -one indicates a ketone; number from the end nearest the C=O group.
- The suffix -ene identifies a double bond; ensure both positions are numbered.
- Always position substituents (like methyl) to achieve the lowest possible numbering.